

## Appendix 2

# Standards and Metal Powder Specifications

| Standard Title                                                                                                                           | Designation |                     |                  |
|------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------|------------------|
|                                                                                                                                          | ASTM        | CIS                 | Other            |
| Materials standards for metal injection-molded parts                                                                                     | ...         | ...                 | MPIF Standard 35 |
| Materials standards for PM self-lubricating bearings                                                                                     | ...         | ...                 | MPIF Standard 35 |
| Materials standards for PM structural parts                                                                                              |             |                     | MPIF Standard 35 |
| Standard for the prevention of fire and dust explosion from the manufacturing, processes, and handling of combustible particulate solids | ...         | GOST 12.1.005-88    | NEFA 654         |
| Standard guide for characterizing properties of metal powders used for additive manufacturing processes                                  | F3049—14    | ...                 | ...              |
| Standard specification for bronze-base powder metallurgy (PM) bearings                                                                   | 438         | ...                 | ...              |
| Standard terminology of powder metallurgy                                                                                                | B243—13     | ...                 | ...              |
| Aluminum pigment, powder and paste, for paint                                                                                            | D 962       | GOST 5494-95        | TT-P-320B        |
| Aluminum powder, flaked, grained, and atomized                                                                                           | ...         | ...                 | MIL-A-512A       |
| Aluminum lumpy pigment                                                                                                                   | ...         | GOST 10096-76       | ...              |
| Antimony, powdered, technical                                                                                                            | ...         | ...                 | MIL-A-10841B     |
| Brass powder (for sintering)                                                                                                             | ...         | ...                 | MIL-B-11552B     |
| Bronze powder, gold                                                                                                                      | D 267       | ...                 | TT-P-340         |
| Copper powder                                                                                                                            | ...         | GOST 4960           | ...              |
| Copper powder (for use in ammunition)                                                                                                    | ...         | ...                 | JAN-C-768        |
| copper electrolytic powder, ultrafine, stabilized                                                                                        | ...         | TU 17911-368-094-96 | ...              |
| Copper reduced powder, stabilized                                                                                                        | ...         | TU 48-O318-O83-94   | ...              |
| Copper powder for use in antifouling paints                                                                                              | D 964       | ...                 | ...              |
| Cobalt powder                                                                                                                            | ...         | GOST 9721-79        | ...              |
| Iridium, powdered                                                                                                                        | ...         | GOST 12338-81       | ...              |
| Magnesium-aluminum alloy, powdered                                                                                                       | ...         | ...                 | JAN-M-454        |
| Magnesium powder (for use in ammunition)                                                                                                 | ...         | ...                 | JAN-M-382A       |
| Manganese, powdered (for use in ammunition)                                                                                              | ...         | ...                 | JAN-M-476A       |
| Nickel powder                                                                                                                            | ...         | GOST 9722-79        | ...              |
| Nickel powder (for use in ammunition)                                                                                                    | ...         | ...                 | JAN-N-412A       |
| Niobium powder                                                                                                                           | ...         | GOST 26252-84       | ...              |

| Standard Title                                                                                                                                                                                                                                                                                                           | Designation |                |                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|----------------|
|                                                                                                                                                                                                                                                                                                                          | ASTM        | CIS            | Other          |
| Palladium, powdered                                                                                                                                                                                                                                                                                                      | ...         | GOST 14836-82  | ...            |
| Powders, metal, atomized (aluminum, magnesium, magnesium-aluminum)                                                                                                                                                                                                                                                       | ...         | ...            | MIL-P-14067B   |
| Rhodium, powdered                                                                                                                                                                                                                                                                                                        | ..          | GOST 12342-81  | ...            |
| Tin powder                                                                                                                                                                                                                                                                                                               | ...         | GOST 9723-79   | ...            |
| Tin, pulverized                                                                                                                                                                                                                                                                                                          | ...         | ...            | JAN-T-458      |
| Titanium, powdered                                                                                                                                                                                                                                                                                                       | ...         | ...            | MIL-T-13405    |
| Tungsten, powdered                                                                                                                                                                                                                                                                                                       | ...         | ...            | MIL-T-13827    |
| Zinc powder                                                                                                                                                                                                                                                                                                              | ...         | GOST 12601-76  | ...            |
| Zinc dust (for use in pyrotechnics)                                                                                                                                                                                                                                                                                      | ...         | ...            | JAN-Z-365      |
| Zinc dust (metallic zinc powder); dry (paint pigment)                                                                                                                                                                                                                                                                    | D 520       | ...            | TT-P-460 No. 1 |
| Zirconium-nickel alloy powder (for use in delay compositions)                                                                                                                                                                                                                                                            | ...         | ...            | MIL-Z—11410A   |
| Zirconium powder (calciothermic)                                                                                                                                                                                                                                                                                         | ...         | TU 48-4-234-84 | ...            |
| Zirconium powder (natriothermic)                                                                                                                                                                                                                                                                                         | ...         | TU 48-4-376-76 | ...            |
| Zirconium, powdered (for use in ammunition)                                                                                                                                                                                                                                                                              | ...         | ...            | JAN-Z-399A     |
| Miscellaneous:                                                                                                                                                                                                                                                                                                           |             |                |                |
| Brushes, electrical contacts, and carbon stock, electrical contact brush                                                                                                                                                                                                                                                 |             |                | MIL-B-003743D  |
| Precision electroformed sieves, specification                                                                                                                                                                                                                                                                            | E 161       | ...            | ...            |
| Standards referenced include the following: American Society for Testing and Materials (ASTM), Commonwealth of Independents States (CIS), MIL—Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402, Metal Powder Industries Federation (MPIF), The National Fire Protection Association. |             |                |                |